

0.2 Sets and Notation

Sets are the foundation of modern mathematics. In linear algebra, we study *vector spaces*, which are sets equipped with operations; *subspaces*, which are subsets with special properties; and *linear maps*, which are functions between sets. This section establishes the notation and operations we will use throughout.

0.2.1 Number Sets

We begin by defining the standard sets of numbers used throughout this book.

Definition 0.2.1 — Common Number Sets

We use the following notation for common sets of numbers:

- $\mathbb{N}$: The set of natural numbers $\{1, 2, 3, \dots\}$.
- $\mathbb{Z}$: The set of integers $\{\dots, -2, -1, 0, 1, 2, \dots\}$.
- $\mathbb{Q}$: The set of rational numbers.
- $\mathbb{R}$: The set of real numbers.
- $\mathbb{C}$: The set of complex numbers.

Definition 0.2.2 — Cartesian Product

Let A and B be sets. The **Cartesian product** of A and B , denoted $A \times B$, is the set of all ordered pairs:

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

More generally, for sets $A_1, A_2, \dots, A_n$, the Cartesian product is:

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) : a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$$

Definition 0.2.3 — n -Tuple

An **n -tuple** is an ordered list of n elements. For a set A , the set of all n -tuples with entries from A is denoted:

$$A^n = \underbrace{A \times A \times \dots \times A}_{n \text{ times}} = \{(a_1, a_2, \dots, a_n) : a_i \in A \text{ for } i = 1, 2, \dots, n\}$$

Example. (n -Tuples)

Here are examples of n -tuples from various sets:

1. $\mathbb{R}^2$: The Cartesian plane. Elements include $(0, 0)$, $(1, 2)$, $(-3, \pi)$, $(\sqrt{2}, -5.7)$.
2. $\mathbb{R}^3$: 3-dimensional space. Elements include $(1, 0, 0)$, $(1, 2, 3)$, $(-1, \pi, e)$.

3. $\mathbb{Z}^2$: Pairs of integers. Elements include $(0, 0)$, $(1, -2)$, $(5, 7)$. Note that $(\frac{1}{2}, 3) \notin \mathbb{Z}^2$.
4. $\mathbb{C}^2$: Pairs of complex numbers. Elements include $(1 + i, 2 - 3i)$, $(i, 0)$, $(3, 4)$.
5. $\{0, 1\}^3$: Binary 3-tuples. This set has exactly 8 elements:

$$\{0, 1\}^3 = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}$$

In general, if A is a finite set with $|A| = k$ elements, then A^n has k^n elements.

0.2.2 Set-Builder Notation and the Empty Set

When defining sets, we often describe them by specifying a property that their elements must satisfy.

Definition 0.2.4 — Set-Builder Notation

Let S be a set and $P(x)$ a predicate. The **set-builder notation**

$$\{x \in S : P(x)\}$$

denotes the set of all elements x in S for which $P(x)$ is true. Variants include $\{x \in S \mid P(x)\}$ and $\{x \in S; P(x)\}$.

Example. (Set-Builder Examples)

The following sets are defined using set-builder notation:

1. $\{x \in \mathbb{R} : x^2 < 4\} = (-2, 2)$, the open interval from -2 to 2 .
2. $\{n \in \mathbb{Z} : n \text{ is even}\} = \{\dots, -4, -2, 0, 2, 4, \dots\}$.
3. $\{(x, y) \in \mathbb{R}^2 : x + y = 1\}$ is the line through $(1, 0)$ and $(0, 1)$.

Definition 0.2.5 — Empty Set

The **empty set**, denoted $\emptyset$ or $\{\}$, is the set containing no elements:

$$\emptyset = \{x : x \neq x\}$$

For any element a , we have $a \notin \emptyset$.

Remark. Statements of the form “ $\forall x \in \emptyset : P(x)$ ” are **vacuously true**—there are no elements to check. This technicality appears when discussing trivial subspaces or the span of an empty collection.

0.2.3 Subsets

The concept of a *subset* is central to linear algebra. When we define a subspace, we first verify it is a subset of the ambient space.

Definition 0.2.6 — Subset

Let A and B be sets. We say A is a **subset** of B , written $A \subseteq B$, if every element of A is also an element of B :

$$A \subseteq B \iff \forall x : (x \in A \Rightarrow x \in B)$$

If $A \subseteq B$ and $A \neq B$, we say A is a **proper subset** of B , written $A \subsetneq B$ or $A \subset B$.

Example. (Subset Examples)

The following illustrate subset relationships:

1. $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$.
2. $\{1, 2\} \subseteq \{1, 2, 3\}$, and $\{1, 2\} \subsetneq \{1, 2, 3\}$.
3. For any set A : $\emptyset \subseteq A$ and $A \subseteq A$.
4. $\{x \in \mathbb{R} : x^2 = 1\} = \{-1, 1\} \subseteq \mathbb{Z}$.

Definition 0.2.7 — Set Equality

Two sets A and B are **equal**, written $A = B$, if they contain exactly the same elements:

$$A = B \iff (A \subseteq B \text{ and } B \subseteq A)$$

Remark. To prove two sets are equal, we typically prove **two subset inclusions**: first show $A \subseteq B$, then show $B \subseteq A$. This technique appears constantly when proving that two subspaces or spans are equal.

0.2.4 Set Operations

Sets can be combined using union, intersection, and difference. These operations appear when discussing sums of subspaces and their intersections.

Definition 0.2.8 — Union

The **union** of sets A and B is the set of elements in A or B (or both):

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

Definition 0.2.9 — Intersection

The **intersection** of sets A and B is the set of elements in both A and B :

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

Two sets are **disjoint** if $A \cap B = \emptyset$.

Definition 0.2.10 — Set Difference

The **set difference** (or **relative complement**) of B in A is:

$$A \setminus B = \{x \in A : x \notin B\}$$

When $B \subseteq U$ for some universal set U , the **complement** of B in U is $B^c = U \setminus B$.

Example. (Set Operations)

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Then:

- $A \cup B = \{1, 2, 3, 4, 5, 6\}$
- $A \cap B = \{3, 4\}$
- $A \setminus B = \{1, 2\}$
- $B \setminus A = \{5, 6\}$

Example. (Set Operations with Intervals)

Let $A = [0, 2]$ and $B = [1, 3]$ be intervals in $\mathbb{R}$. Then:

- $A \cup B = [0, 3]$
- $A \cap B = [1, 2]$
- $A \setminus B = [0, 1)$

Remark. In linear algebra, we will see that the **intersection** of two subspaces is always a subspace, but the **union** of two subspaces is generally *not* a subspace (this is a common exam question!). Instead of union, we use the **sum** of subspaces: $U + W = \{u + w : u \in U, w \in W\}$.